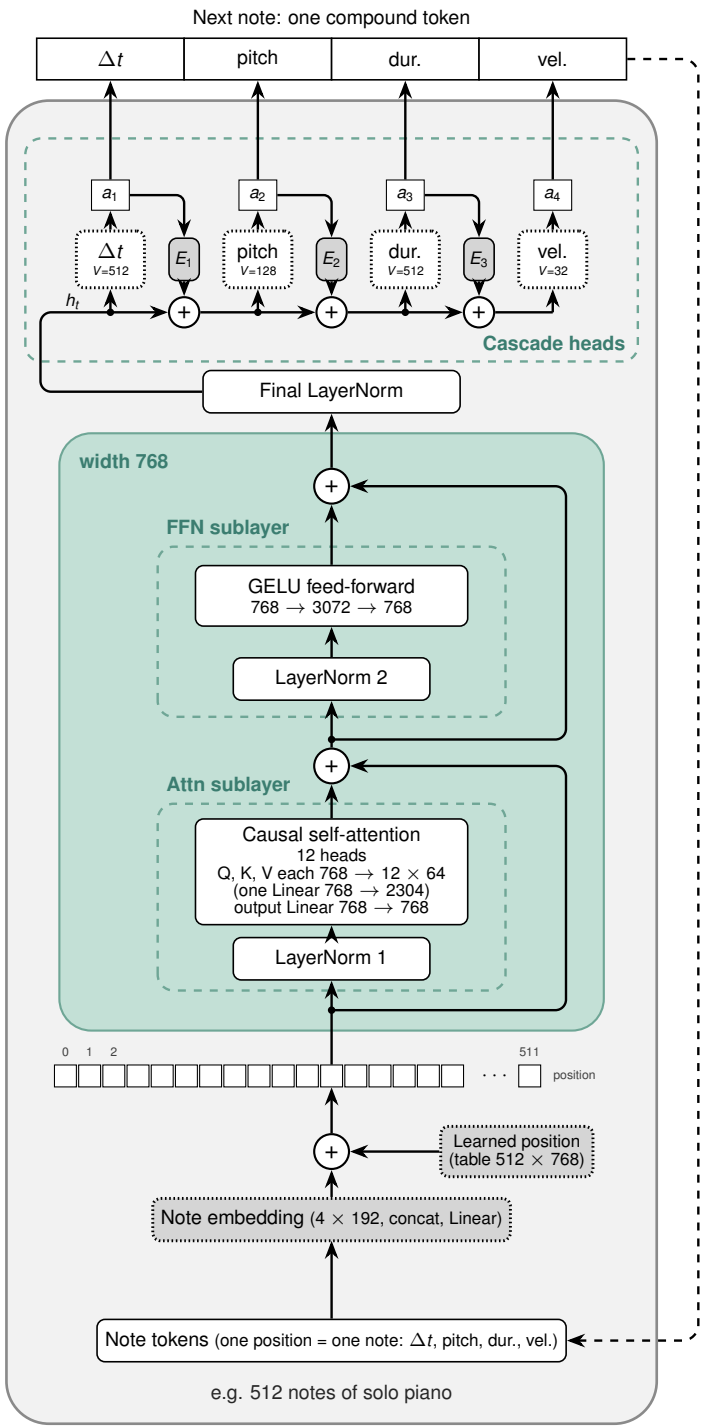
musicar-S-piano (107M)

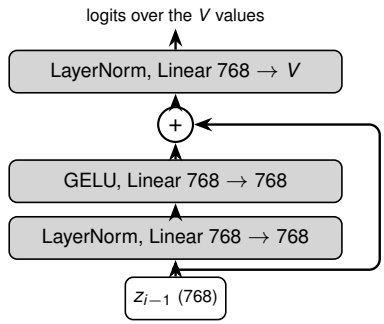
106.84M parameters: trunk 99.11M, cascade heads 6.52M, embeddings 1.21M. Solo piano, context 512 notes.

How to read: numbers give the reading order; teal = the repeated block; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part.

1 Music model



3 One cascade head



Why a cascade. Head i (a dotted box, own weights) reads the trunk state plus the embeddings E of the values already chosen,

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

so the note's probability follows the chain rule: $p(\Delta t)\, p(\text{pitch} \mid \Delta t) \cdots$. In training a_i is the true value (teacher forcing), in generation the sampled one. Loss: the sum of the four cross-entropies.

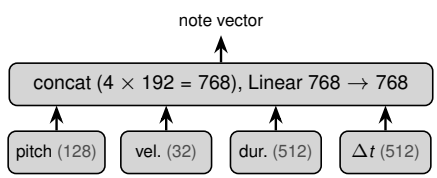
4 Training (done, 2026-09-30)

Data: MAESTRO + GiantMIDI-Piano (≈5% of the notes) and Aria-MIDI (≈95%), solo piano.

Run: 38,000 steps × 61,440 notes (120 windows of 512) = 2.33B notes, ≈4 passes; Muon + AdamW, LR 10^{-3}, WSD schedule; 12 h 27 min on a laptop RTX 5060 (8 GB).

Result: Aria-MIDI validation 5.706 nats / note, inside the 5.47–5.72 forecast of the scaling ladder; MAESTRO + GiantMIDI-Piano validation 6.987.

2 Input embedding (the dotted boxes)



+ learned position: a table of 512 vectors, one per place in the window, added to the note vector; so the model reads at most 512 notes.

Time is absolute: Δt and durations in 20 ms steps, clipped at 10.22 s. No bars, beats or tempo tokens; no instrument, no start / end tokens, no window conditions.